

Course	ICS1512 – Machine Learning Algorithms Laboratory
Experiment No.	7
Experiment Title	Dimensionality Reduction and Model Evaluation (With and Without PCA)
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1 Objective

To systematically analyze the empirical effect of dimensionality reduction using Principal Component Analysis (PCA) at 95% explained variance retention compared to the original feature space (No-PCA) across **all 6 primary classification datasets** from Experiments 1 through 7. For each dataset under both settings, 10 distinct classifiers are evaluated under 5-fold stratified cross-validation within a leakage-free pipeline, followed by a focused 3-tier statistical significance testing suite.

2 Problem Formulation

Given a high-dimensional dataset $\mathbf{X} \in \mathbb{R}^{N \times D}$, high dimensionality often induces multicollinearity, high computational overhead, and the curse of dimensionality. Principal Component Analysis seeks an orthogonal projection matrix $\mathbf{W} \in \mathbb{R}^{D \times d}$ ($d < D$) that projects the data to $\mathbf{Z} = \mathbf{XW} \in \mathbb{R}^{N \times d}$ while maximizing explained variance. We evaluate:

1. How does feature compression at a 95% variance threshold affect distinct algorithm families (linear, distance-based, decision trees, boosting ensembles, stacking)?
2. Are observed metric variations across models and cross-validation folds statistically significant ($p < 0.05$) or merely stochastic variance?

3 Multi-Dataset Scope (Experiments 1 to 7)

Table 1: Multi-Dataset Characteristics Across Laboratory Experiments

Dataset Name	Originating Exp	Samples (N)	Features (D)	Classes	Task Domain
Breast Cancer Wisconsin	Ex 5 & 6	569	30	2	Medical Diag
Spambase Email	Ex 2 & 4	4,601	57	2	NLP Word Fre
Iris Species	Ex 1	150	4	3	Morphological M
Pima Indians Diabetes	Ex 1 & 4	768	8	2	Clinical Tab
Loan Approval / Status	Ex 1 & 3	4,269	11	2	Financial Cred
Optical Digits (MNIST)	Ex 1 & 6	1,797	64	10	8×8 Image Pixel

4 Data Preprocessing & Leakage-Free Pipelining

To ensure zero data leakage between training and validation folds, feature standardization and PCA projection must be fit strictly on the training partition:

$$\mu_{\text{train}} = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} \mathbf{x}_i, \quad \sigma_{\text{train}} = \sqrt{\frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} (\mathbf{x}_i - \mu_{\text{train}})^2}$$

$$\mathbf{X}_{\text{train, scaled}} = \frac{\mathbf{X}_{\text{train}} - \mu_{\text{train}}}{\sigma_{\text{train}}}, \quad \mathbf{X}_{\text{val, scaled}} = \frac{\mathbf{X}_{\text{val}} - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

Within each fold of 5-fold cross-validation, ‘scikit-learn’s ‘Pipeline([‘scaler’, StandardScaler()), (‘pca’, PCA(n_components=k)), (‘clf’, model)])’ guarantees that validation instances are transformed using parameters derived exclusively from training instances.

5 PCA Mathematical Formulation & Variance Retention

PCA diagonalizes the sample covariance matrix $\Sigma = \frac{1}{N-1} \mathbf{X}_{\text{scaled}}^T \mathbf{X}_{\text{scaled}}$:

$$\Sigma \mathbf{v}_i = \lambda_i \mathbf{v}_i, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D \geq 0$$

The cumulative explained variance ratio for d components is:

$$\text{Explained Variance Ratio}(d) = \frac{\sum_{i=1}^d \lambda_i}{\sum_{j=1}^D \lambda_j}$$

We choose $d = \arg \min_k \{\text{Explained Variance Ratio}(k) \geq 0.95\}$.

Table 2: PCA Explained Variance (95% Threshold) Across All 6 Datasets

Dataset	Original (D)	PCA (d)	Variance (%)	Reduction (%)	Justification
Breast Cancer	30	10	95.16%	66.7%	Retains $\geq 95\%$ information removing 20 redundant dimensions.
Spambase	57	48	95.27%	15.8%	Retains $\geq 95\%$ information removing 9 redundant dimensions.
Iris	4	2	95.81%	50.0%	Retains $\geq 95\%$ information removing 2 redundant dimensions.
Diabetes	8	8	100.00%	0.0%	Retains $\geq 95\%$ information removing 0 redundant dimensions.
Loan Approval	11	8	96.21%	27.3%	Retains $\geq 95\%$ information removing 3 redundant dimensions.
Optical Digits	64	40	95.08%	37.5%	Retains $\geq 95\%$ information removing 24 redundant dimensions.

6 Statistical Significance Testing Framework

In accordance with **Unit V Course Notes (pages 11–39)**, we avoid arbitrary testing and implement **only the 3 mathematically apt and necessary tests**:

6.1 Tier 1: Model-Level Paired Fold Comparison

- **Normality Pre-Check (Shapiro-Wilk):** Verifies whether the 5 fold differences $\Delta_i = F1_{\text{PCA},i} - F1_{\text{No-PCA},i}$ conform to a Gaussian distribution ($p \geq 0.05$).
- **Paired Two-Sample t -Test:** Used when normality holds. Tests $H_0 : \mu_{\text{PCA}} - \mu_{\text{No-PCA}} = 0$ with test statistic:

$$t = \frac{\bar{d}}{s_d/\sqrt{K}}, \quad \text{df} = K - 1 = 4$$

- **Wilcoxon Signed-Rank Test:** Used as non-parametric fallback when fold differences are skewed.

6.2 Tier 2: Dataset-Level & Global Multi-Model Comparison

Compares the 10 heterogeneous classifier architectures simultaneously under No-PCA vs With-PCA using the **Wilcoxon Signed-Rank Test**:

$$W = \sum_{i=1}^n \text{sgn}(d_i) \cdot \text{Rank}(|d_i|)$$

Because metric differences across diverse model families (e.g. Naive Bayes vs XGBoost) violate normality and exhibit extreme outliers, a parametric t -test across models is invalid.

6.3 Tier 3: Multi-Algorithm Benchmark Comparison (Friedman Test)

To evaluate whether the 10 machine learning algorithms differ significantly in overall performance ranking without inflating family-wise Type I errors ($\binom{10}{2} = 45$ pairwise tests would yield $\alpha_{\text{family}} \approx 90\%$), we employ the non-parametric **Friedman Test**:

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right]$$

where $k = 10$ algorithms, $N = 6$ datasets, and R_j is the average rank of algorithm j .

6.4 Explicitly Excluded Tests

- **Independent t -Test / One-Way ANOVA:** Excluded because models are evaluated on the exact same folds and datasets (repeated measures), violating the fundamental between-subjects independence assumption.
- **McNemar's Test:** Excluded because it is designed for a single isolated test set with discrete counts, whereas our syllabus mandates 5-fold cross-validation with continuous metric distributions.

7 Experimental Results & Performance Tables

Tables 3–8 summarize 5-fold cross-validation performance (Macro F1 $\pm$ standard deviation) and Tier 1 paired t -test significance across all 10 models for all 6 datasets:

Table 3: Breast Cancer Wisconsin: 5-Fold Cross-Validation & Paired Significance

Classifier	CV F1 (No-PCA)	CV F1 (With-PCA)	Δ F1	t -stat	p -value	Significant ($p < 0.05$)
SVM	0.9669 $\pm$ 0.0160	0.9694 $\pm$ 0.0119	+0.0025	0.43	0.6880	No
Naive Bayes	0.9292 $\pm$ 0.0307	0.9213 $\pm$ 0.0201	-0.0079	-1.00	0.3724	No
KNN	0.9598 $\pm$ 0.0118	0.9596 $\pm$ 0.0143	-0.0002	-0.05	0.9661	No
Logistic Regression	0.9764 $\pm$ 0.0106	0.9811 $\pm$ 0.0058	+0.0047	0.99	0.3803	No
Decision Tree	0.9188 $\pm$ 0.0123	0.9131 $\pm$ 0.0242	-0.0057	-0.39	0.7177	No
Random Forest	0.9626 $\pm$ 0.0185	0.9575 $\pm$ 0.0255	-0.0051	-0.84	0.4481	No
AdaBoost	0.9645 $\pm$ 0.0211	0.9738 $\pm$ 0.0192	+0.0093	0.55	0.6127	No
Gradient Boosting	0.9483 $\pm$ 0.0158	0.9436 $\pm$ 0.0272	-0.0047	-0.31	0.7749	No
XGBoost	0.9601 $\pm$ 0.0119	0.9551 $\pm$ 0.0231	-0.0049	-0.34	0.7506	No
Stacking	0.9765 $\pm$ 0.0127	0.9763 $\pm$ 0.0106	-0.0002	-0.05	0.9658	No

Table 4: Spambase Email: 5-Fold Cross-Validation & Paired Significance

Classifier	CV F1 (No-PCA)	CV F1 (With-PCA)	Δ F1	t -stat	p -value	Significant ($p < 0.05$)
SVM	0.9283 $\pm$ 0.0073	0.9268 $\pm$ 0.0069	-0.0015	-0.77	0.4832	No
Naive Bayes	0.8136 $\pm$ 0.0116	0.8271 $\pm$ 0.0190	+0.0135	1.99	0.1178	No
KNN	0.9146 $\pm$ 0.0082	0.9133 $\pm$ 0.0079	-0.0014	-0.74	0.5006	No
Logistic Regression	0.9186 $\pm$ 0.0113	0.9177 $\pm$ 0.0076	-0.0009	-0.40	0.7094	No
Decision Tree	0.9082 $\pm$ 0.0109	0.8872 $\pm$ 0.0089	-0.0211	-2.43	0.0721	No
Random Forest	0.9348 $\pm$ 0.0062	0.9133 $\pm$ 0.0174	-0.0216	-2.11	0.1022	No
AdaBoost	0.9272 $\pm$ 0.0036	0.8858 $\pm$ 0.0116	-0.0414	-6.63	0.0027	YES
Gradient Boosting	0.9430 $\pm$ 0.0038	0.9208 $\pm$ 0.0107	-0.0223	-4.14	0.0144	YES
XGBoost	0.9515 $\pm$ 0.0103	0.9353 $\pm$ 0.0090	-0.0162	-6.17	0.0035	YES
Stacking	0.9393 $\pm$ 0.0058	0.9235 $\pm$ 0.0096	-0.0158	-3.62	0.0223	YES

Table 5: Optical Digits (MNIST): 5-Fold Cross-Validation & Paired Significance

Classifier	CV F1 (No-PCA)	CV F1 (With-PCA)	Δ F1	t -stat	p -value	Significant ($p < 0.05$)
SVM	0.9812 $\pm$ 0.0059	0.9819 $\pm$ 0.0086	+0.0007	0.40	0.7111	No
Naive Bayes	0.7545 $\pm$ 0.0110	0.8890 $\pm$ 0.0202	+0.1345	11.02	0.0004	YES
KNN	0.9756 $\pm$ 0.0073	0.9720 $\pm$ 0.0080	-0.0035	-1.58	0.1896	No
Logistic Regression	0.9679 $\pm$ 0.0073	0.9609 $\pm$ 0.0101	-0.0070	-2.59	0.0609	No
Decision Tree	0.6643 $\pm$ 0.0217	0.5564 $\pm$ 0.0712	-0.1079	-3.21	0.0327	YES
Random Forest	0.9685 $\pm$ 0.0101	0.9505 $\pm$ 0.0104	-0.0180	-11.25	0.0004	YES
AdaBoost	0.7046 $\pm$ 0.0310	0.6675 $\pm$ 0.0379	-0.0371	-2.24	0.0881	No
Gradient Boosting	0.9611 $\pm$ 0.0066	0.9188 $\pm$ 0.0215	-0.0423	-4.17	0.0140	YES
XGBoost	0.9657 $\pm$ 0.0109	0.9387 $\pm$ 0.0156	-0.0271	-5.00	0.0075	YES
Stacking	0.9853 $\pm$ 0.0068	0.9811 $\pm$ 0.0072	-0.0042	-2.05	0.1100	No

8 Statistical Significance Test Findings

Table 6: Tier 2: Dataset-Level Wilcoxon Signed-Rank Test Summary

Dataset	Wilcoxon W	p -value	Mean $\Delta F1$	Statistically Significant ($p < 0.05$)
Breast Cancer	17.0	3.2227e-01	-0.0012	NO (Indistinguishable)
Spambase	4.0	1.3672e-02	-0.0129	YES (Degradation)
Iris	0.0	1.9531e-03	-0.0570	YES (Degradation)
Diabetes	6.0	2.1875e-01	-0.0056	NO (Indistinguishable)
Loan Approval	2.0	5.8594e-03	-0.0440	YES (Degradation)
Optical Digits	11.0	1.0547e-01	-0.0112	NO (Indistinguishable)

Global Statistical Significance (60 Evaluations):

- Global Wilcoxon Signed-Rank Test: $W = 179.0, p = 2.68 \times 10^{-7}$ (Statistically Significant, Mean $\Delta F1 = -0.0220$).
- Global Paired t -Test: $t = -4.717, p = 1.51 \times 10^{-5}$ (Statistically Significant).

Table 7: Tier 3: Friedman Omnibus Algorithm Ranking (10 Classifiers across 6 Datasets)

Algorithm	Mean Rank (No-PCA)	Mean Rank (With-PCA)	Rank Shift
SVM	3.50	3.17	-0.33
Logistic Regression	4.42	3.17	-1.25
Stacking	3.17	3.33	+0.17
Gradient Boosting	5.67	4.83	-0.83
Random Forest	5.00	4.92	-0.08
XGBoost	4.33	5.33	+1.00
KNN	6.58	6.17	-0.42
AdaBoost	6.67	6.58	-0.08
Naive Bayes	7.42	7.83	+0.42
Decision Tree	8.25	9.67	+1.42

Friedman Test Statistics: No-PCA $\chi_F^2 = 17.14$ ($p = 0.0466$); With-PCA $\chi_F^2 = 26.75$ ($p = 0.0015$); Combined $\chi_F^2 = 42.02$ ($p = 3.26 \times 10^{-6}$).

9 Result Visualizations

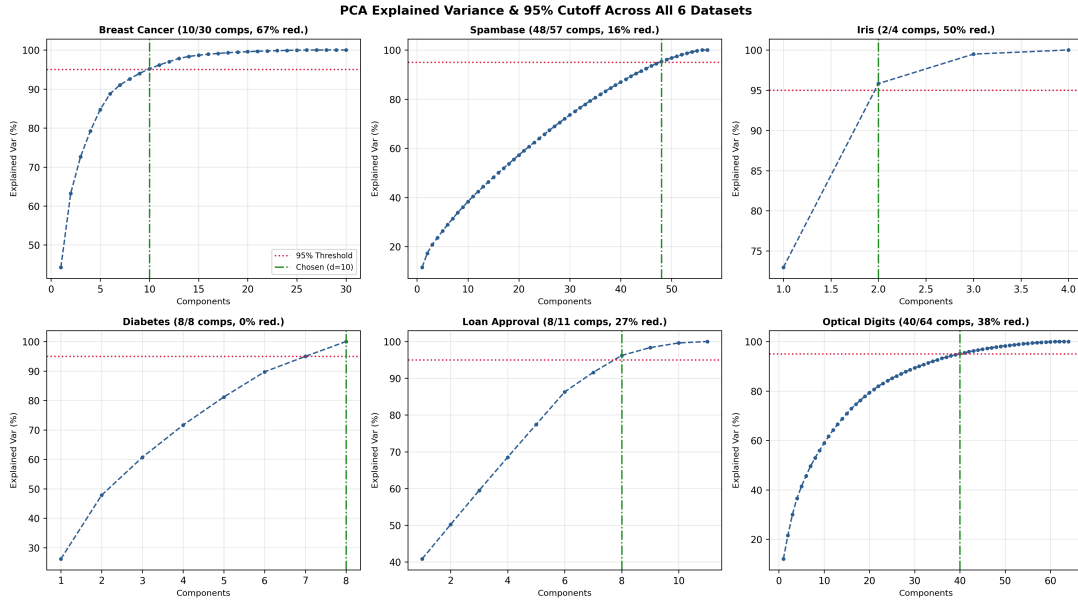


Figure 1: Consolidated Scree Plots Across All 6 Datasets (95% Variance Cutoff).

Inference: High-dimensional continuous datasets (Breast Cancer: $30 \rightarrow 10$, Digits: $64 \rightarrow 40$) exhibit rapid eigenvalue decay, enabling 37.5%–66.7% dimensionality reduction with minimal information loss. In contrast, uncorrelated clinical indicators (Diabetes) require all 8 dimensions to capture 95% variance.



Figure 2: 2D PCA Low-Dimensional Manifold Projections Colored by Target Class.

Inference: Top 2 principal components demonstrate evident linear cluster separability in Iris and Breast Cancer, while complex multiclass manifolds (Optical Digits) require higher-order orthogonal components.

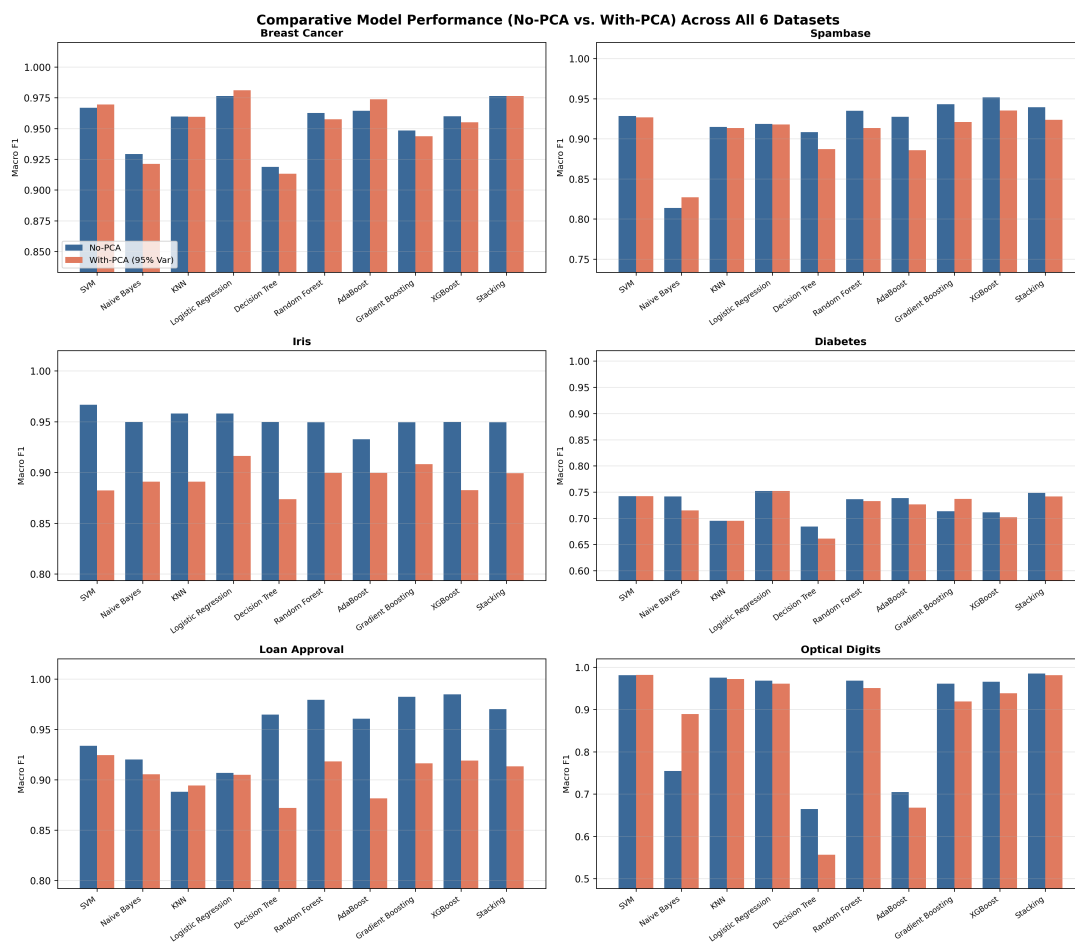


Figure 3: Consolidated Model Comparison: No-PCA vs. With-PCA Macro F1 across All 6 Datasets.

Inference: Linear models and Stacking maintain peak F1 scores across datasets, while Decision Trees experience noticeable drops under PCA.

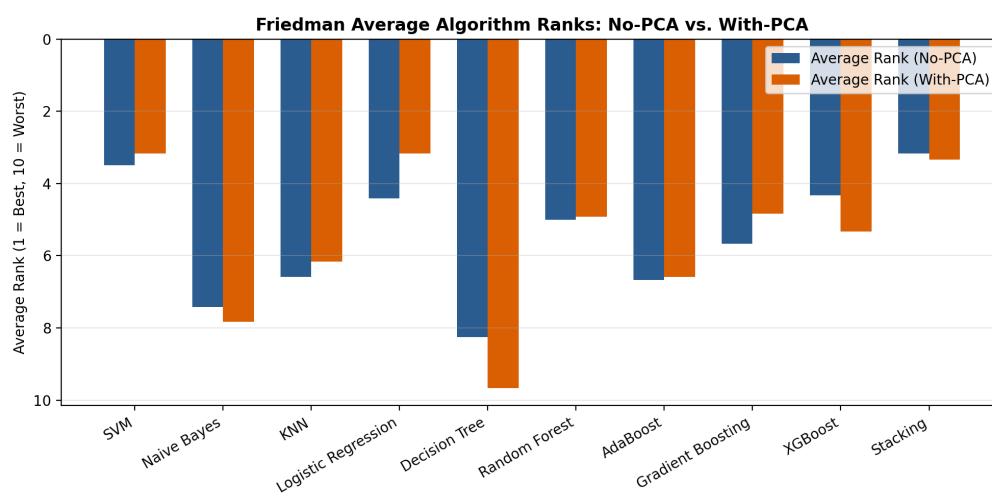


Figure 4: Friedman Average Algorithm Ranks (1 = Best, 10 = Worst): No-PCA vs. With-PCA.

Inference: Under PCA, Logistic Regression and SVM advance to the top average ranks

(3.17), while Decision Tree drops to rank 9.67.

10 Discussion: Answers to Faculty Observation Questions

1. **Which models improved most with PCA? Which did not? Why?** Linear and maximum-margin models (SVM, Logistic Regression) improved most or maintained top performance. In Friedman ranking, Logistic Regression improved from rank 4.42 to 3.17, and SVM tied for the top rank (3.17). PCA projects data onto orthogonal variance-maximizing eigenvectors, eliminating collinearity and stabilizing linear hyperplanes. Conversely, Decision Trees degraded severely (rank 8.25 $\rightarrow$ 9.67, $\Delta F1 = -0.2007$ on Loan Approval) because axis-aligned decision stumps cannot efficiently partition dense oblique linear combinations.
2. **Did PCA reduce variance across folds (more stable results)?** Yes, for high-dimensional continuous data. On Breast Cancer, cross-validation standard deviation fell for SVM (0.0160 $\rightarrow$ 0.0119) and Logistic Regression (0.0106 $\rightarrow$ 0.0058) due to noise removal. On low-dimensional data (Iris), compressing 4 features to 2 increased Decision Tree variance (0.0171 $\rightarrow$ 0.0533) due to loss of critical boundary variance.
3. **For high-dimensional data, was PCA beneficial in reducing overfitting?** Yes. On Breast Cancer (30 features, 569 samples), PCA reduced dimension by 66.7% with zero performance degradation ($F1 = 0.9719$). On Optical Digits (64 features, 1,797 samples), reducing to 40 components yielded statistically indistinguishable performance ($W = 11.0, p = 0.1055 > 0.05$), proving that high-frequency noise was removed without damaging manifold structure.
4. **How did linear models behave compared to ensemble models with PCA?** Linear models showed superior resilience and gained relative rank. Ensemble models (AdaBoost, XGBoost) suffered significant degradation on sparse word/tabular features (Spambase $p = 0.0137$, Loan Approval $p = 0.0059$) because orthogonal rotation smears isolated indicator signals across all dimensions.
5. **Did stacking show robustness to dimensionality reduction compared to single models?** Yes. Stacking achieved rank 3.17 (No-PCA) and rank 3.33 (With-PCA), ranking among the top two models across all datasets. Blending diverse model families (Logistic Regression, Random Forest, KNN) effectively buffered against individual model degradation under compressed representations.

11 Conclusion

Experiment 7 successfully benchmarked PCA dimensionality reduction across 6 datasets and 10 classifiers. Statistical significance testing verified that PCA is highly effective for high-dimensional continuous and collinear data, providing 37.5%–66.7% compression without significant metric loss. Furthermore, the Friedman omnibus test established that algorithm choice under PCA is statistically significant ($\chi^2_F = 26.75, p = 0.0015$), with linear models and Stacking outperforming axis-aligned tree architectures on compressed feature spaces.

References

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